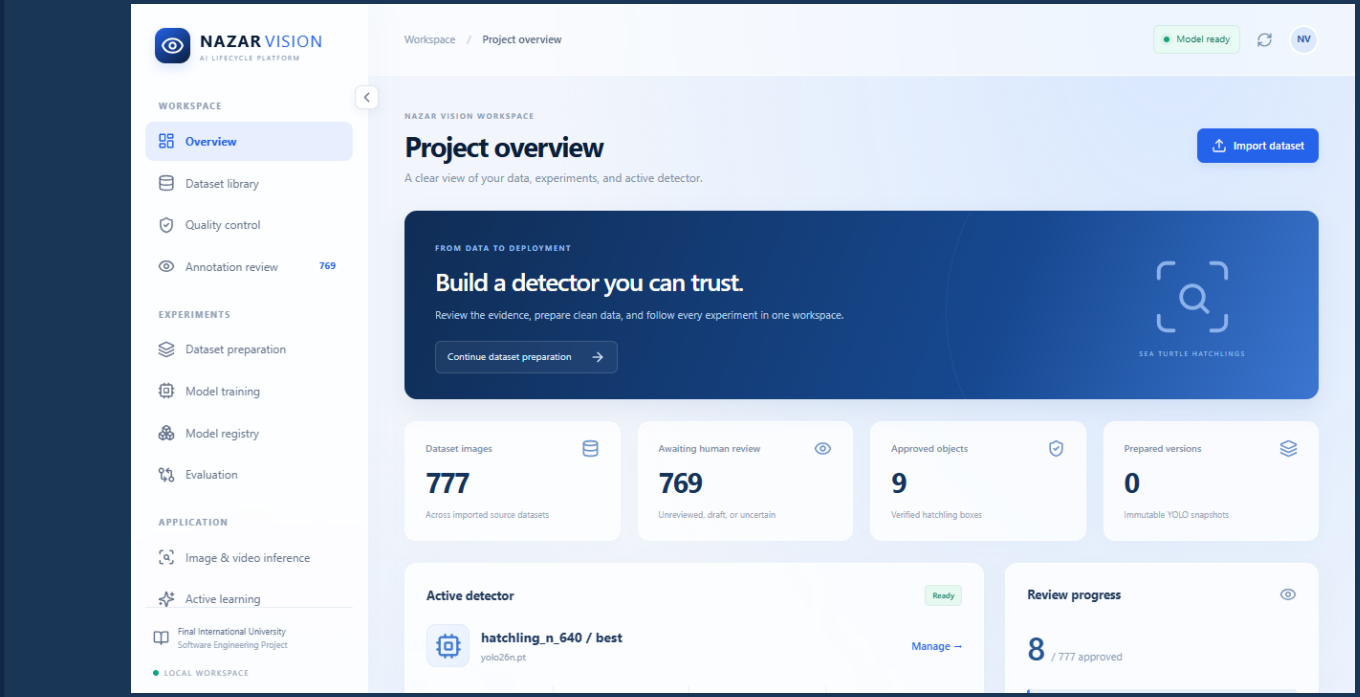
MODEL LAB / SCREEN-RECORDING REFERENCE

Hatchling detector Model Workspace Guide

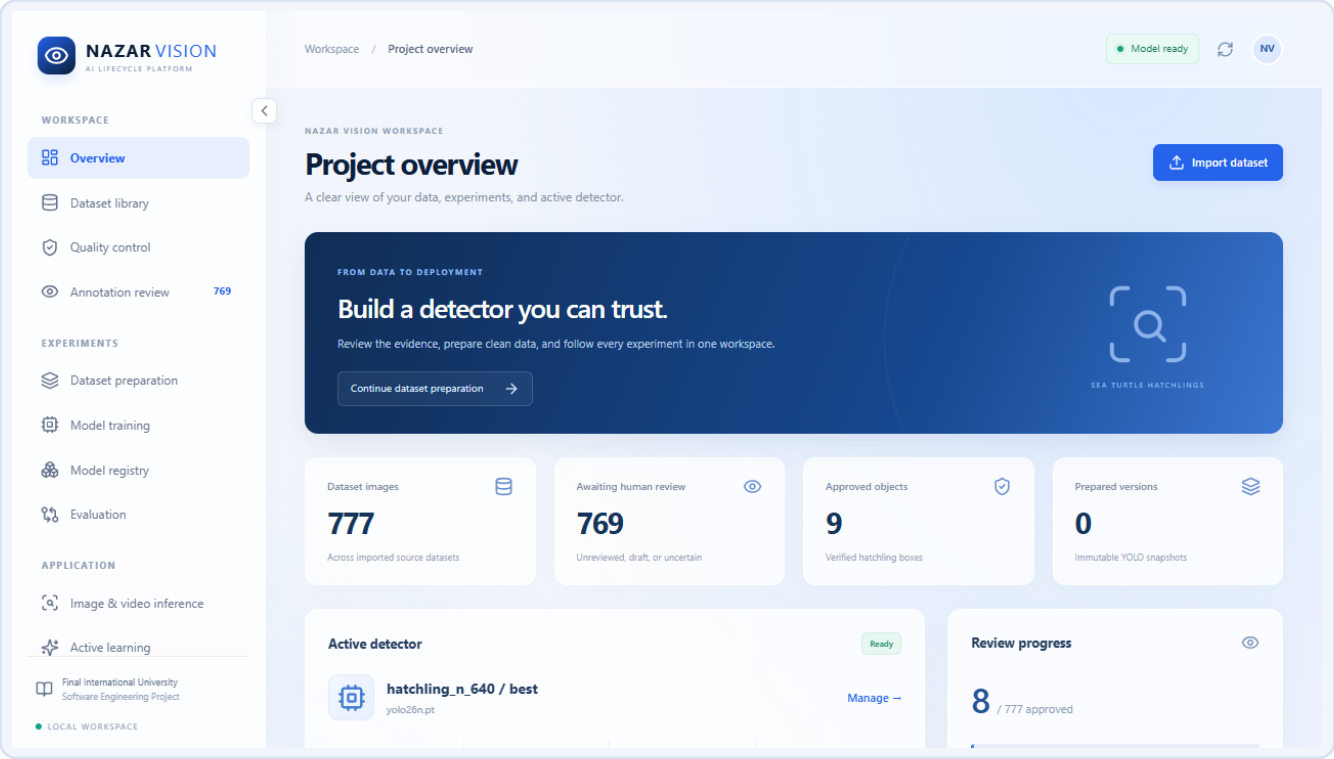
A screenshot, purpose, best practice, and narration cue for every section.

This is a dedicated local workbench for developing and visualizing the hatchling detector. The model is intended to serve as the brain behind Nazar Vision. This workspace is the model lab for that component, not the full product.

SOURCE DATA > HUMAN REVIEW > PREPARED VERSION > TRAINING > EVALUATION > INFERENCE



Overview



WHAT IT IS FOR

A snapshot of dataset volume, review progress, approved boxes, prepared versions, the active detector, recorded metrics, compute, and recent jobs.

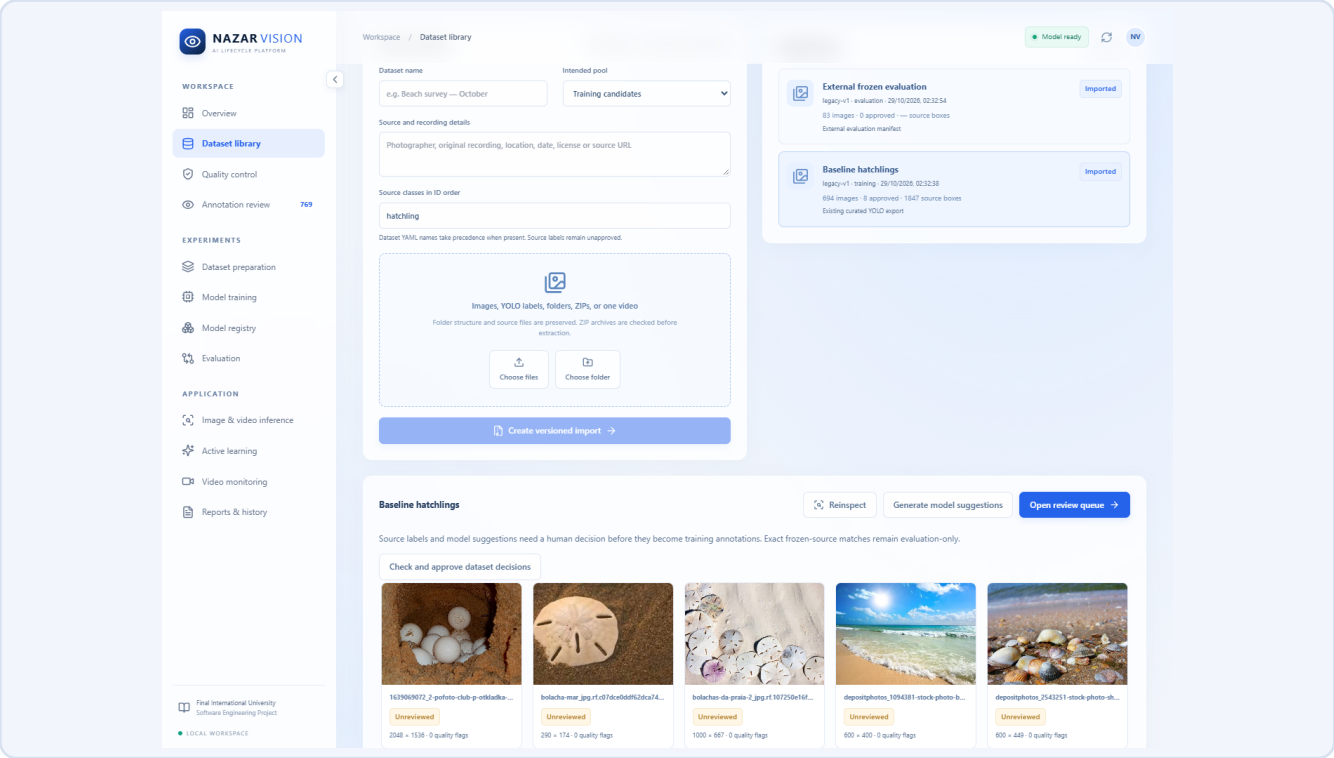
BEST PRACTICE

Use it to decide what needs attention next. Open the underlying dataset, model, or job before drawing conclusions from a total or metric. Read the metric scope shown beside the active model.

SCREEN-RECORDING CUE

Start by identifying the active detector and whether its validation evidence is recorded. Counts describe workflow state; they are not model-quality scores.

Dataset library



WHAT IT IS FOR

Imports and catalogs source images, folders, archives, and videos with their source details, classes, intended pool, and provenance.

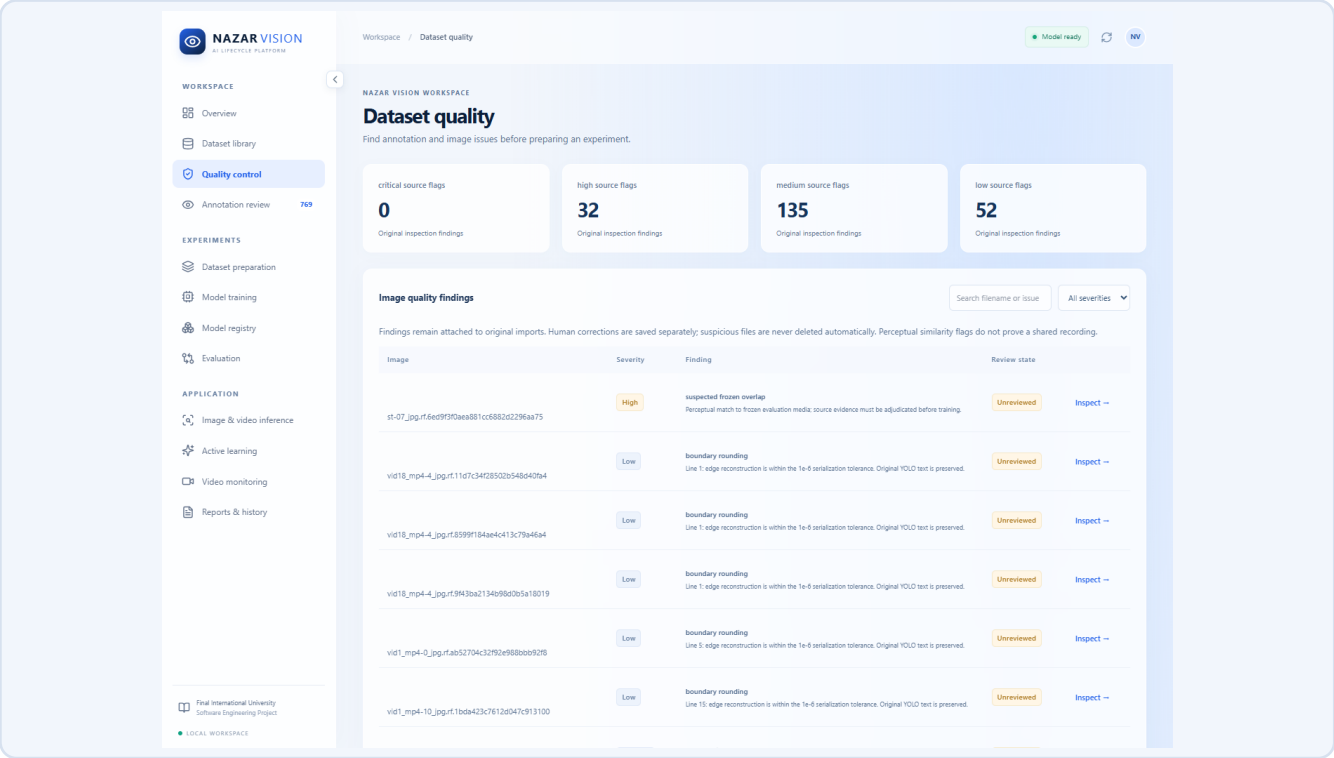
BEST PRACTICE

Preserve original media and record where it came from. Keep training candidates separate from sources reserved for frozen evaluation. Avoid merging unrelated sources just to make a larger dataset.

SCREEN-RECORDING CUE

Open a dataset and show its source and intended pool. An import makes media available for review; existing labels are not automatically verified training truth.

Quality control



WHAT IT IS FOR

Surfaces unreadable files, malformed labels, invalid boxes, duplicates, overlaps, and source-grouping concerns before preparation.

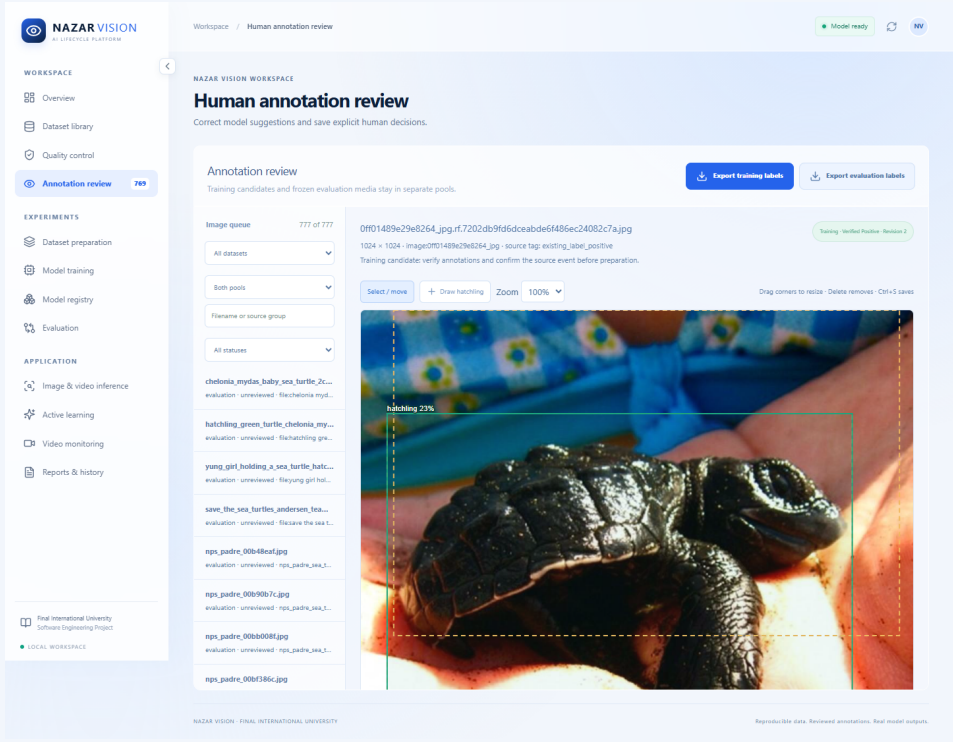
BEST PRACTICE

Investigate each finding against the original media. Treat flags as prompts for review, not automatic delete instructions. Resolve confirmed file and label errors before creating a training version.

SCREEN-RECORDING CUE

Open one representative finding and compare it with the source. Explain what evidence would justify correcting, excluding, or retaining the example.

Annotation review



WHAT IT IS FOR

Lets a human inspect images, correct hatchling boxes, review model suggestions, and save explicit positive, negative, uncertain, rejected, or excluded decisions.

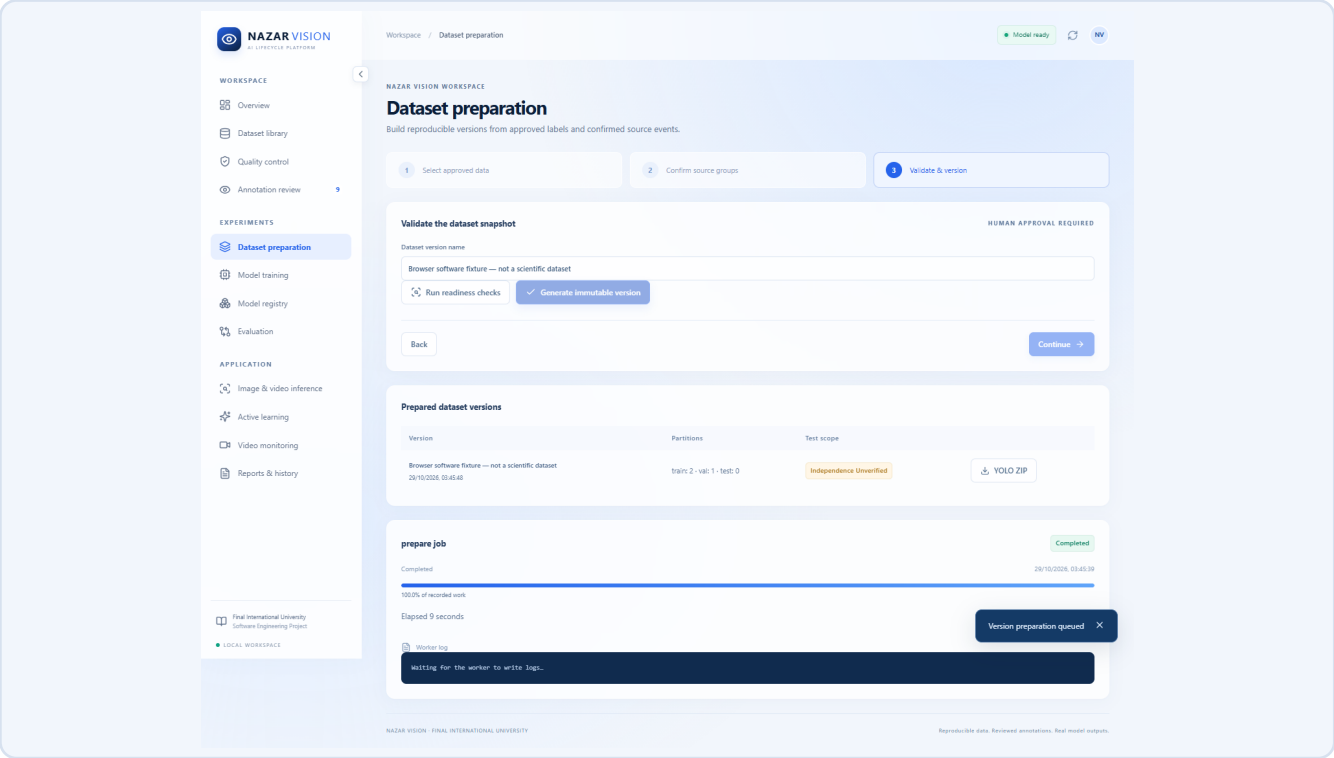
BEST PRACTICE

Inspect the whole image and every box. Approve positive only when hatchlings have valid boxes; approve negative only when none are present. Keep uncertain cases uncertain and record reviewer/evidence notes. Suggestions are not ground truth until reviewed.

SCREEN-RECORDING CUE

Zoom in on one example, inspect or correct a box, then show how the saved decision records the human review.

Dataset preparation



WHAT IT IS FOR

Turns selected approved reviews into a versioned YOLO dataset with source groups, train/validation/test partitions, checksums, and integrity findings.

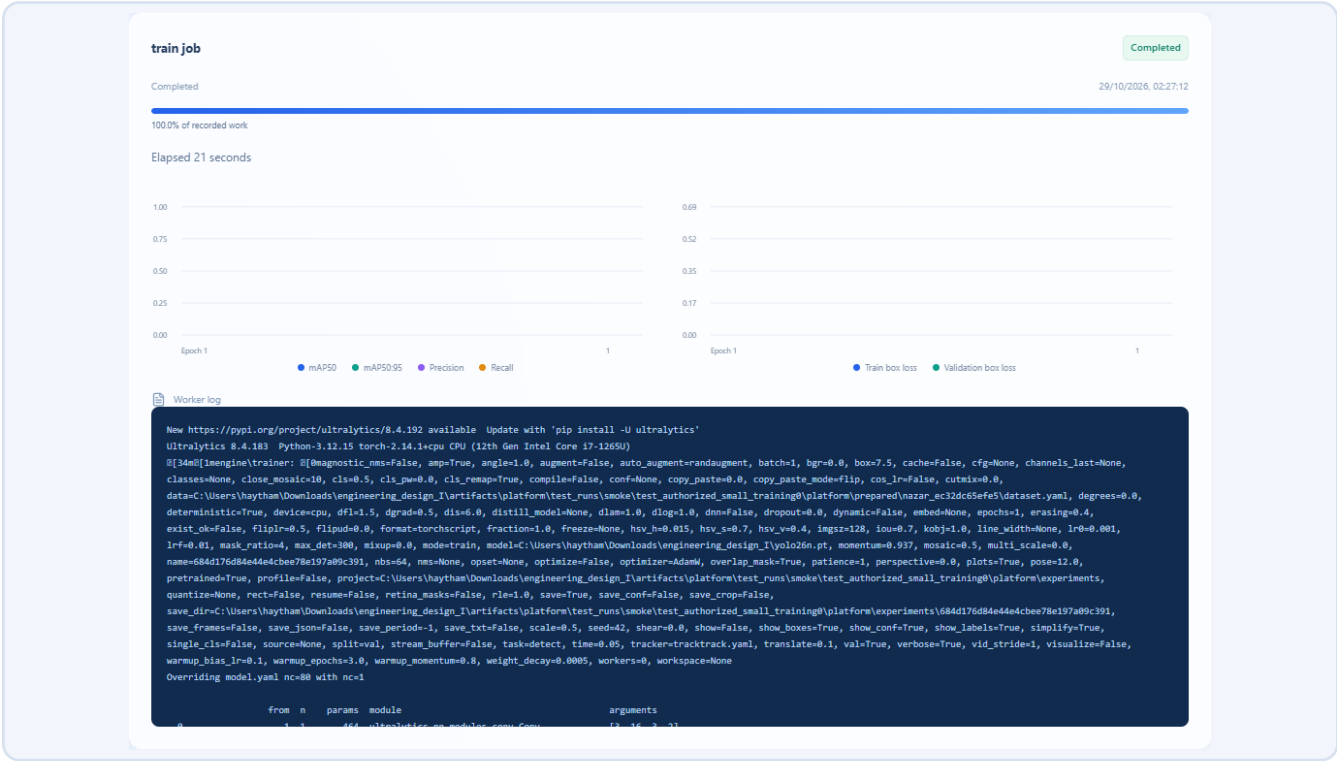
BEST PRACTICE

Group media from the same recording or wildlife event before assigning partitions so related frames do not leak across train, validation, and test. Keep frozen evaluation sources out of training. Read every integrity finding.

SCREEN-RECORDING CUE

Trace a sample to its source group and partition. Explain that the prepared version is a reproducible snapshot used by a training run.

Model training



WHAT IT IS FOR

Starts a bounded experiment from a prepared version and initialization checkpoint, recording settings, resources, logs, epoch progress, metrics, and checkpoints.

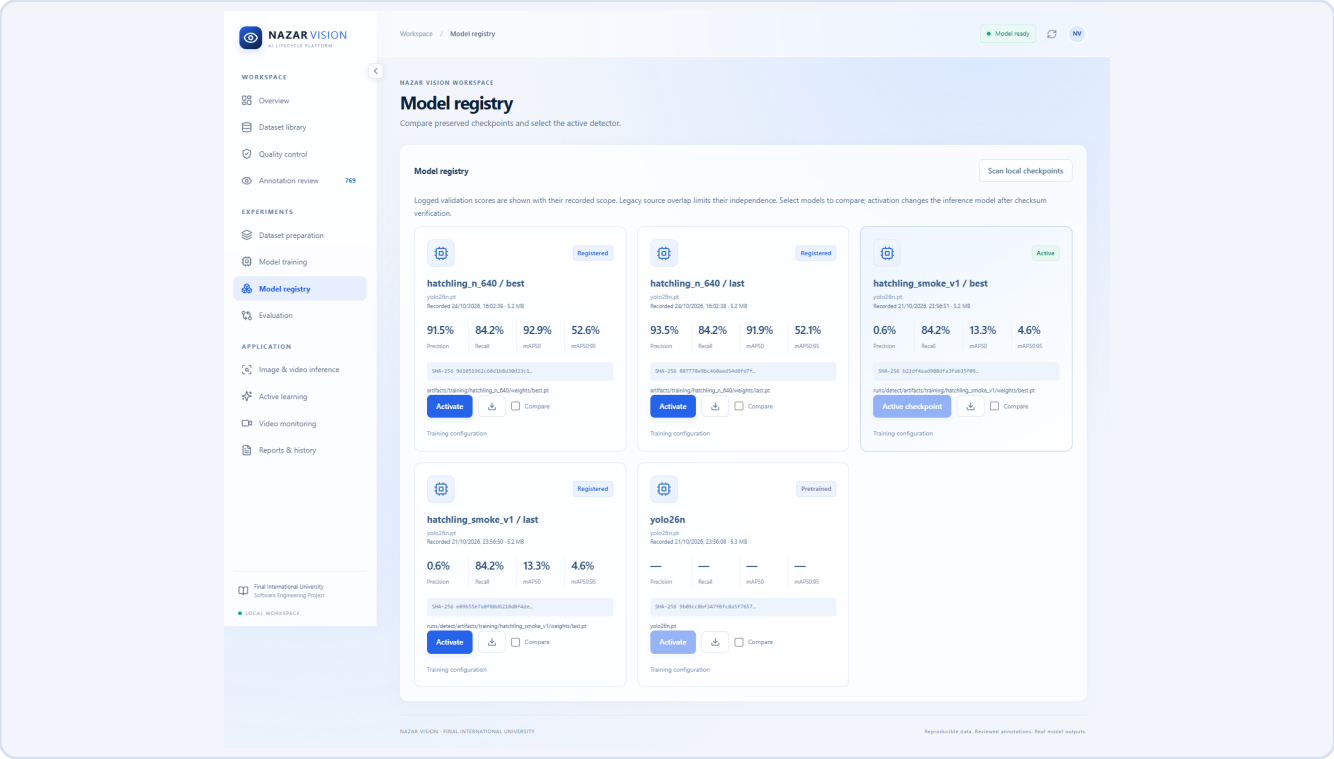
BEST PRACTICE

Check the dataset, checkpoint, device, image size, epochs, batch, learning rate, optimizer, seed, and resource limits before starting. Change settings intentionally, keep each run's configuration, preserve the baseline, and do not describe incomplete jobs as finished results.

SCREEN-RECORDING CUE

Prefer an existing completed run for the recording. Show its configuration and real epoch history. Start a new job only when you intend to use the compute and wait for its result.

Model registry



WHAT IT IS FOR

Lists preserved checkpoints with their training configuration, file size, checksum, recorded metrics, and active status. Supports comparison and activation.

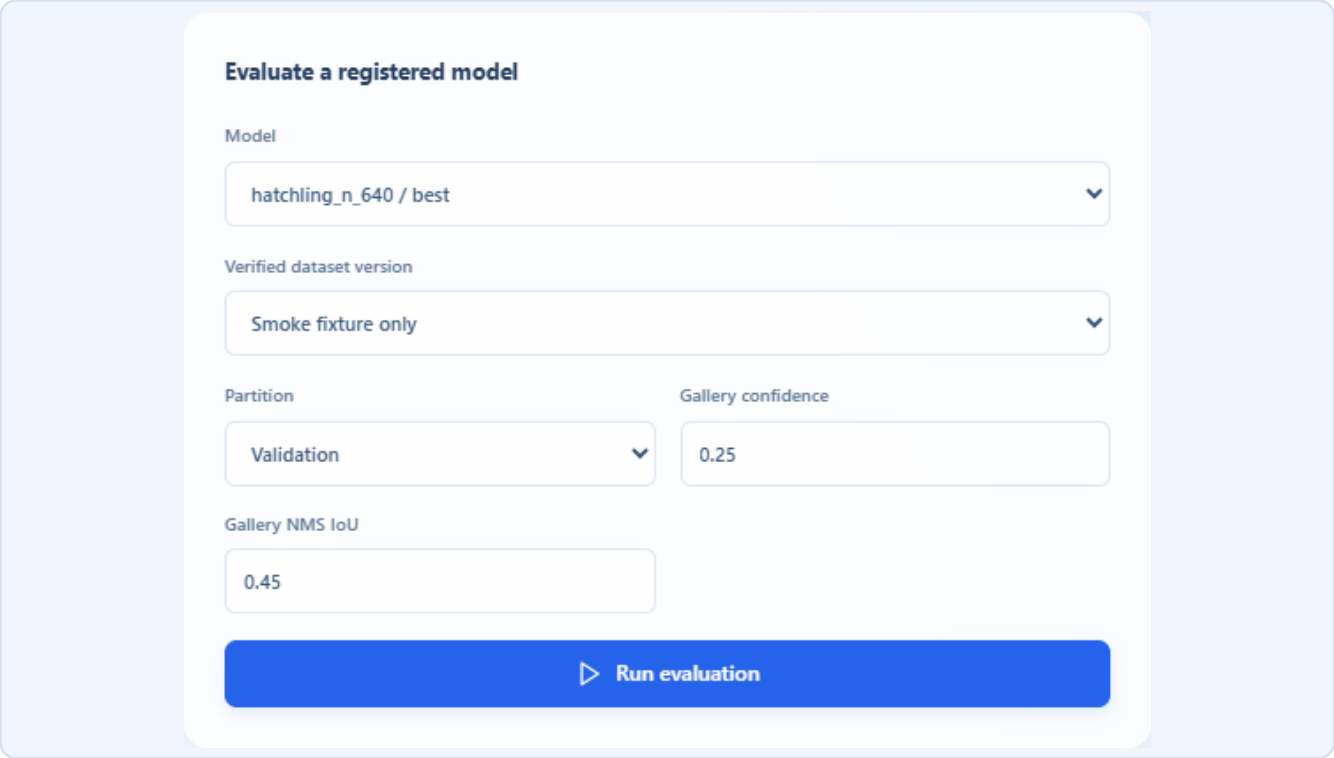
BEST PRACTICE

Compare models only when their datasets and evaluation scopes are comparable. Check model name and checksum before activation. Keep initialization weights distinct from a trained detector and retain the baseline.

SCREEN-RECORDING CUE

Open a trained checkpoint, identify its source run and dataset version, then show how the active model is selected and verified.

Evaluation



WHAT IT IS FOR

Scores a trained model against a selected dataset partition and stores summary metrics, plots, and examples of misses and false detections.

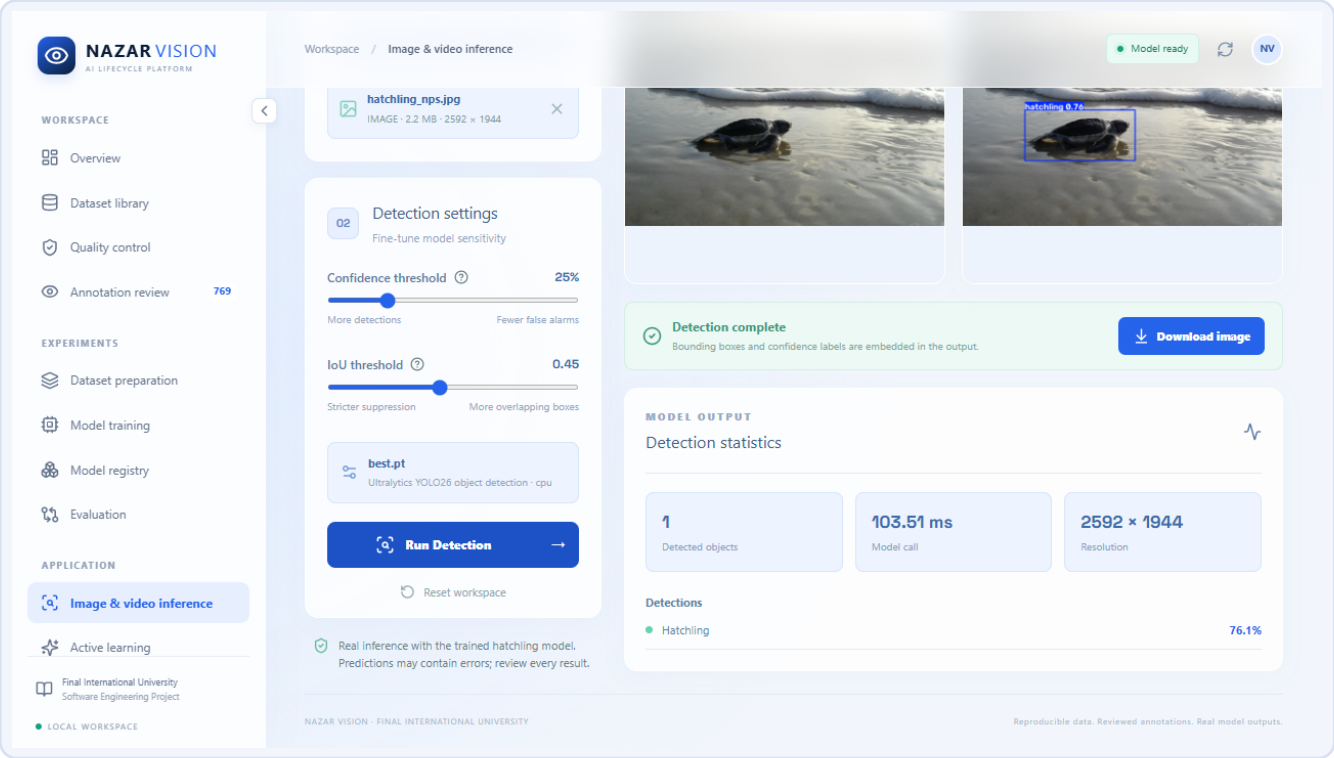
BEST PRACTICE

Use validation data during development. Keep held-out test media out of training and model selection. Review precision, recall, mAP, sample size, source independence, and the error gallery together. Interpret every metric within its recorded scope.

SCREEN-RECORDING CUE

Name the model, dataset version, and partition before reading results. Open a miss or false positive to connect aggregate metrics to model behavior.

Image & video inference



WHAT IT IS FOR

Runs the active detector on uploaded media and shows actual boxes, confidence scores, and downloadable annotated results.

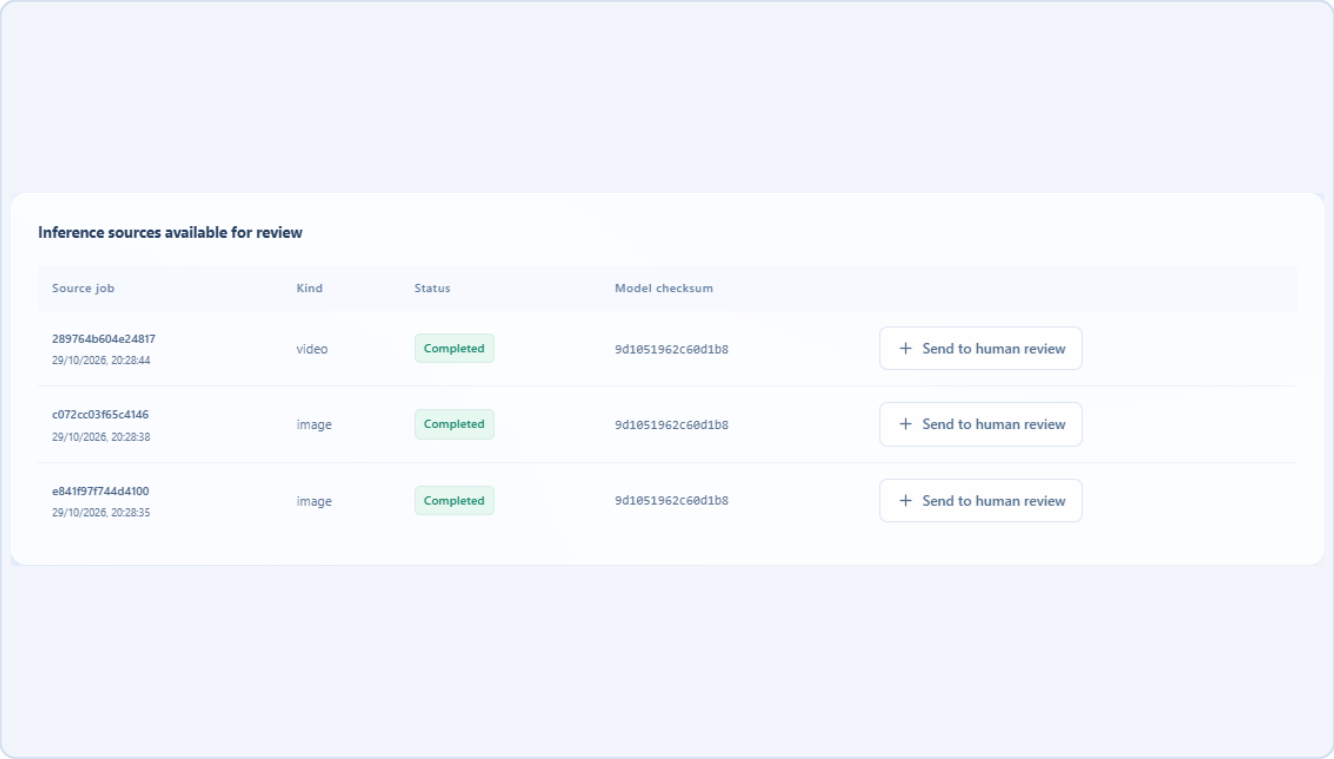
BEST PRACTICE

Confirm the active checkpoint and state the confidence and IoU settings. Inspect original and result. Zero boxes means no detections under those settings, not proof of absence. Video boxes are frame-level detections, not unique animals.

SCREEN-RECORDING CUE

Run a representative image or short clip, point out a result and a limitation, and read the settings visible on screen.

Active learning



WHAT IT IS FOR

Sends difficult or uncertain inference examples into a candidate review flow before any data is prepared for a later experiment.

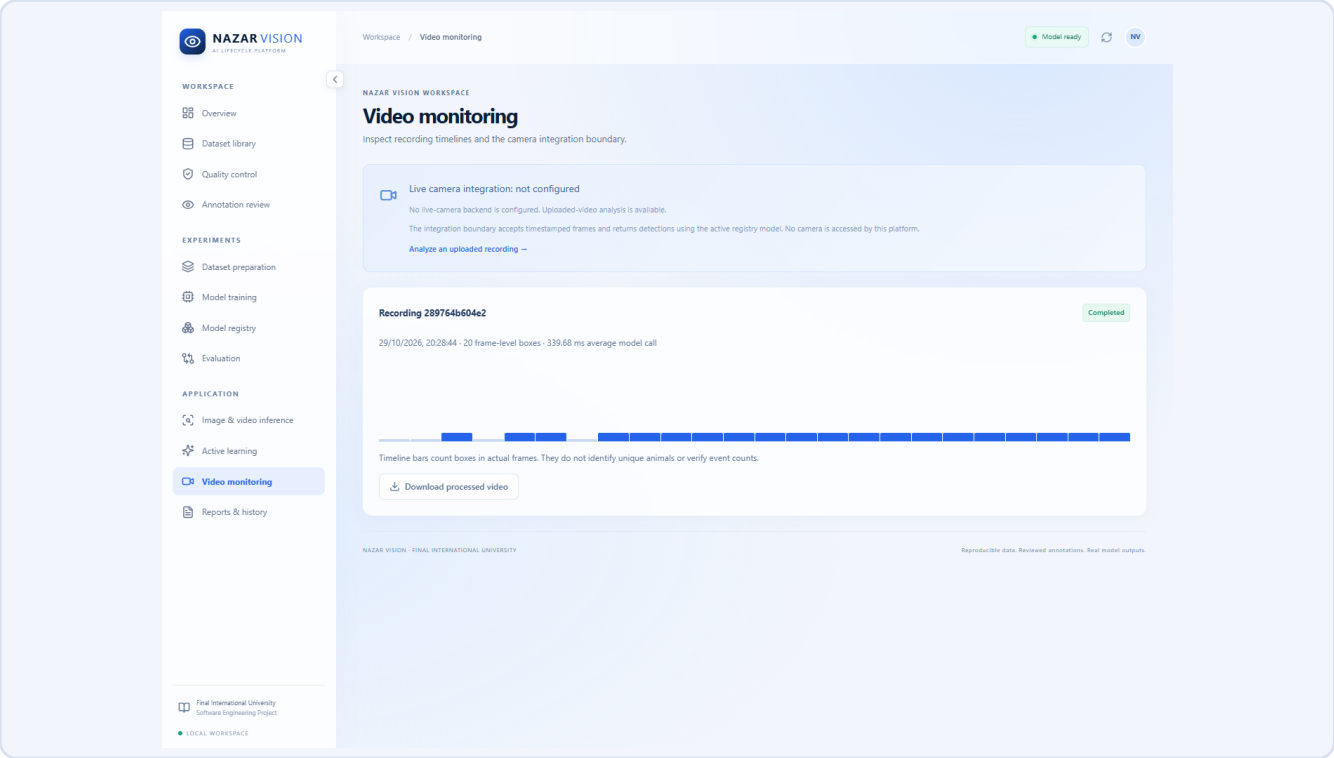
BEST PRACTICE

Prioritize diverse, informative examples instead of near-duplicates. Preserve source and event information. Human-review every candidate before it enters an approved dataset version.

SCREEN-RECORDING CUE

Trace an uncertain inference back to the review queue. The model proposes where to look; the reviewer decides what is true.

Video monitoring



WHAT IT IS FOR

Displays processed recording timelines and frame-level detections, plus the boundary of the current camera integration.

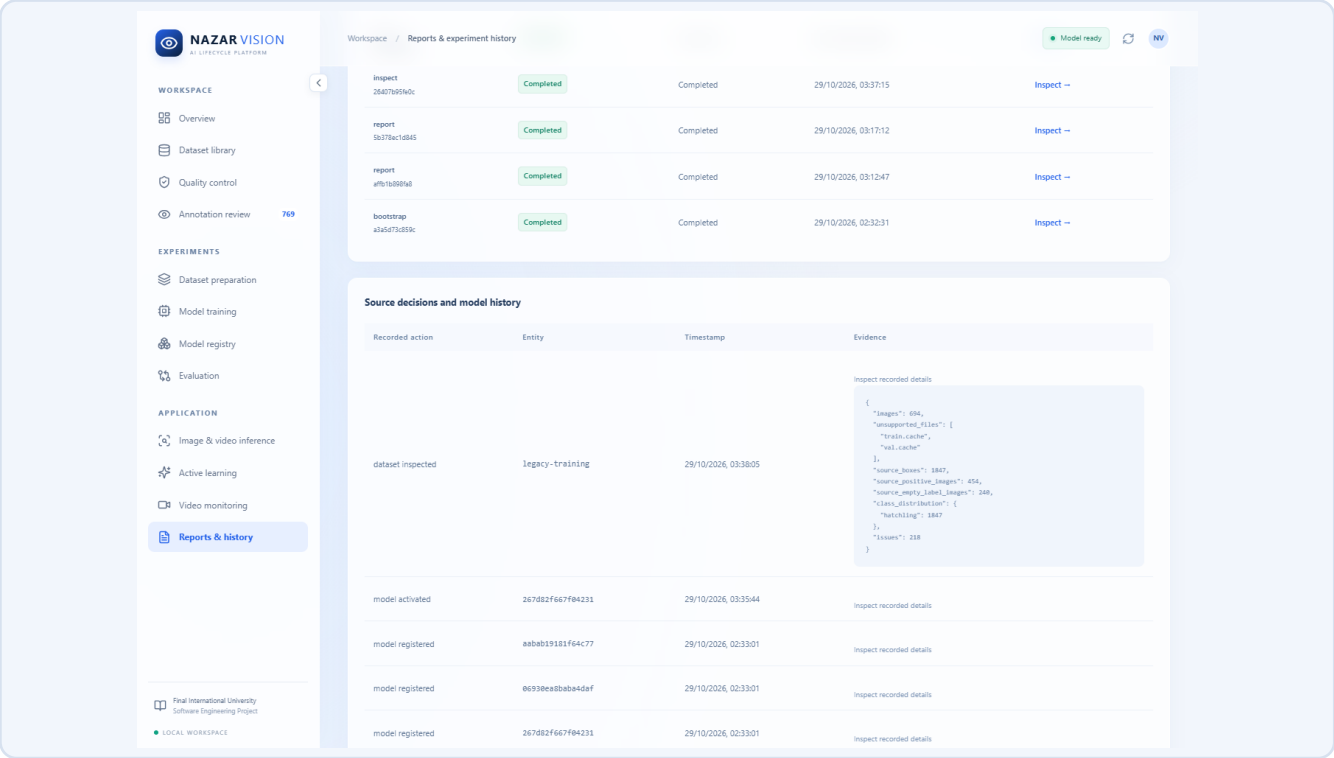
BEST PRACTICE

Describe this as inspection of uploaded recordings. Live-camera integration is not configured, and frame detections do not count unique animals. Keep source video and processing settings available.

SCREEN-RECORDING CUE

Open a processed recording and point to detections over time. State clearly that this is not a live camera connection or individual turtle tracking.

Reports & history



WHAT IT IS FOR

Provides downloadable technical reports and history of dataset, review, preparation, training, model, evaluation, and inference activity.

BEST PRACTICE

Keep each conclusion traceable to the dataset version, checkpoint checksum, settings, partition, job ID, and date. Use reports to preserve evidence; do not rely on a cropped screenshot without run context.

SCREEN-RECORDING CUE

Open one completed job or report and trace its model and data inputs. Close by showing how the experiment can be revisited later.

A clear end-to-end recording

FOLLOW THE MODEL LIFECYCLE

1. Overview: identify the active detector and current workflow state.
2. Dataset library and Quality control: show source provenance and how issues are found.
3. Annotation review: demonstrate the human-in-the-loop step.
4. Dataset preparation: show event grouping and the immutable dataset version.
5. Model training: open a completed run and explain its recorded settings.
6. Model registry and Evaluation: connect the checkpoint to evidence and visible errors.
7. Image & video inference and Active learning: show outputs and route difficult cases back for review.
8. Video monitoring and Reports & history: explain limits and preserve traceability.

KEEP THE CLAIMS ACCURATE

Read metrics with their displayed dataset and evaluation scope; captured values can differ in a fresh workspace.

A small software fixture is not an accuracy benchmark. Independence requires source-event evidence and overlap checks.

Frame-level boxes are not unique-animal counts. The current monitoring page does not connect to a live camera.

Model suggestions remain candidates until a person reviews and saves a decision.